

APURV PATEL

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SUMMARY

Software engineer with 5+ years of experience building production systems and ML infrastructure at Capital One. Designed payments infrastructure serving 1M+ customers, built a data pipeline platform across 10+ teams, and a RAG system with hybrid retrieval and LLM integration. AWS Certified Solutions Architect.

WORK EXPERIENCE

Capital One, Toronto, ON

Principal Software Engineer (Jan 2025 – Present) | **Senior Software Engineer** (Jan 2023 – Dec 2024) | **Associate Software Engineer** (Sept 2021 – Dec 2022)

- Designed and delivered near-real-time payments system using event-driven architecture (Kafka, SQS, EventBridge) across 5+ teams including vendor APIs, mobile, and enterprise platform reducing wait time from 2 days to 15 minutes for 1M+ customers
- Replaced a third-party vendor by building an automated decisioning pipeline (PySpark, AWS Glue) that processes 500–1,000+ daily requests using data-driven classification logic with end-to-end observability via CloudWatch and Splunk
- Built a centralized data pipeline platform on Apache Airflow (ECS Fargate, RDS, Redis/Celery), onboarding ~40 pipelines across 10+ teams with ~70% RTE reduction — created reusable DAG templates and runbooks adopted org-wide
- Migrated 40+ data pipelines to PySpark, creating the reference implementation adopted across Canada, and tuned Spark jobs on EMR for fault tolerance with checkpoint/retry logic and memory optimization
- Created an automated code campaign tool that generated 6,000+ PRs in a single quarter for security updates, replacing ~15 min of manual work per repo — adopted by the enterprise platform team as a built-in feature
- Built a fraud-prevention microservice (Python, Lambda, DynamoDB) for post-approval device verification, serving ~30k customers/month and enabling instant virtual card provisioning

Cyphercor Inc., Ottawa, ON

Software Development Engineer (Jan 2021 – Sept 2021)

- Built 10+ REST APIs and monitoring dashboards for enterprise MFA solutions using Java, React, and SQLite, serving security-sensitive clients across cloud and on-premise deployments
- Owned end-to-end development and CI/CD pipeline for 4 projects, with >90% test coverage across unit and integration tests

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, SQL, Go

AI/ML: LangChain, LangGraph, RAG Pipelines, Vector Databases (pgvector), Prompt Engineering, LLM Evaluation, Scikit-learn, PyTorch

Cloud: AWS (Lambda, ECS, EMR, Glue, S3, EventBridge), GCP, Docker, Terraform

Data & Streaming: Apache Airflow, Kafka, PySpark, Celery, Redis, PostgreSQL, DynamoDB

Tools & Observability: Splunk, CloudWatch, New Relic, Git, Jenkins, CI/CD, pytest, JUnit

PROJECTS

[ap-finrag](#) — RAG system for SEC financial filings with hybrid retrieval (pgvector + Postgres FTS via Reciprocal Rank Fusion), LangChain, streaming SSE with citation tracking, and provider-agnostic LLM integration

[ap-feature-store](#) — ML feature store with offline/online serving, sub-millisecond Redis inference, point-in-time correct retrieval, batch materialization, and REST API serving

[ap-streambeat](#) — Real-time CDN quality monitor with Kafka ingestion, Isolation Forest anomaly detection, atomic model hot-swap, and WebSocket alerting

EDUCATION

University of Waterloo — MEng, Electrical and Computer Engineering (GPA: 3.95/4.0) | Dec 2020

Gujarat Technological University — BEng, Computer Engineering (GPA: 9.03/10) | June 2019

CERTIFICATIONS

AWS Certified Solutions Architect – Associate | Capital One Machine Learning Engineer Training Program (2024)